

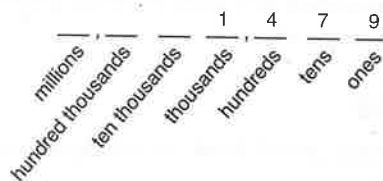
Number Sense and Problem Solving

Lesson 1: Compare and Order Numbers

Place Value

Numbers are part of your everyday life. Whether you're paying with cash, reading bus schedules, or changing television channels, you're using whole numbers.

The value of a number depends on the **place value** of its digits. On the place-value chart below, note that the value of a digit increases as you move to the left.



In the whole number shown on the chart, 1 has the greatest value. The number would be read as "one thousand four hundred seventy-nine."

Comparing and Ordering Values

To **order** numbers, you need to compare the value of their digits. Align the place values of the numbers you are comparing. Start at the left and compare the value of the first digit of each number.

Example 1: Place the numbers 342, 98, and 317 in order from least to greatest.

1. Compare the first digit in each number. Since 98 is a two-digit number, and the other numbers are three-digit numbers, 98 has the smallest value.
2. Next, compare the first digit of the remaining numbers. Since both numbers have the same digit (3) in the hundreds place, compare the digits in the tens place. Since 1 is less than 4, 317 is less than 342.

From least to greatest:
98, 317, 342

Rounding Numbers

Place value is also essential to **rounding** numbers.

Example 2: Round 2451 to the nearest hundred.

1. Locate the place value that you want to round to. Then look at the place value to the right. If the digit to the right is 5 or greater, round up. If the digit to the right is less than 5, don't change the number in the hundreds place.
2. Round up the digit in the hundreds place. Then change the digits to the right of the hundreds place to zeros.

The value to the right of the hundreds place is a 5. Round up.

2451 rounds up to **2500**

Key Ideas

- When ordering numbers, compare the same place values between numbers.
- Round numbers to estimate answers or eliminate answer choices in multiple-choice questions.

GED® TEST TIP

It's important to know the inequality symbols below.

$<$ means "is less than"

$\leq$ means "is less than or equal to"

$>$ means "is greater than"

$\geq$ means "is greater than or equal to"

Practice 1

- A. Write the digit from the number below that corresponds to the listed place value. The first one is done for you.

1,436,879

- | | |
|-----------------------------------|-------------------------------|
| <u> 4 </u> 1. hundred thousands | <u> </u> 5. thousands |
| <u> </u> 2. hundreds | <u> </u> 6. tens |
| <u> </u> 3. ones | <u> </u> 7. ten thousands |
| <u> </u> 4. millions | |

- B. Round these numbers as directed.

- | | |
|--|---|
| 8. Round 544 to the nearest hundred. | 11. Round 11,632 to the nearest thousand. |
| 9. Round 76 to the nearest ten. | 12. Round 1525 to the nearest thousand. |
| 10. Round 1058 to the nearest hundred. | 13. Round 84 to the nearest hundred. |

- C. In each of the following pairs, which number is greater?

- | | |
|----------------|------------------------|
| 14. 100 or 89 | 16. 1099 or 1145 |
| 15. 339 or 341 | 17. 125,391 or 119,450 |

- D. Write these numbers in order from least to greatest.

- | | |
|-----------------------|--------------------------|
| 18. 23 18 45 39 | 21. 1429 1420 1432 1425 |
| 19. 111 89 109 91 | 22. 12,071 11,098 12,131 |
| 20. 1087 932 909 1139 | 23. 15,356 15,309 15,298 |

- E. Choose the one best answer to each question.

- | | |
|--|---|
| 24. When stacking items, the heaviest items should be placed at the bottom. Starting at the bottom, in what order should items weighing 45 pounds, 40 pounds, 50 pounds, and 48 pounds be stacked?
A. 40, 45, 48, 50
B. 45, 50, 40, 48
C. 50, 48, 45, 40
D. 50, 40, 45, 48 | 25. Which of the following correctly shows 1,543,976 rounded to the nearest hundred thousand?
A. 2,000,000
B. 1,600,000
C. 1,500,000
D. 1,540,000 |
|--|---|

Answers and explanations begin on page 661.

Lesson 2: Whole Number Operations

Key Ideas

- Add when combining amounts and subtract when finding the difference.
- Align place-value columns when adding or subtracting.
- Always look to the next place-value column to regroup.

GED® TEST TIP

Another way to check answers is to round the numbers in a problem and find an approximate answer.

Addition and Subtraction

You use **addition** when you need to combine amounts. The answer in an addition problem is called the **sum**, or **total**. When you are adding, it's helpful to stack the numbers in a column. Be sure to line up the place-value columns and then work from right to left, starting with the ones column.

Sometimes the digits in a place-value column add up to 10 or more. When this happens, you will need to **regroup** to the next place value.

Example 1: Add $40 + 129 + 24$.

1. Align the numbers you want to add on the ones column. Working from right to left, add the ones column first. Since the ones column totals 13, write the 3 in the ones column and regroup, or **carry**, the 1 ten to the tens column.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 3 \end{array}$$

2. Add the tens column, including the regrouped 1.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 93 \end{array}$$

3. Then add the hundreds column. Since there is only one value, write the 1 hundred in the answer.

$$\begin{array}{r} 1 \\ 40 \\ 129 \\ + 24 \\ \hline 193 \end{array}$$

You **subtract** when you want to find the **difference** between amounts. Write the greater number on top, and align the amounts on the ones column. You may also need to regroup as you subtract.

Example 2: If Sue is 57 and Kathy is 38, how many years older is Sue?

1. Find the difference in their ages. Start with the ones column. Since 7 is less than the number being subtracted (8), regroup, or **borrow**, 1 ten from the tens column. Add the regrouped amount to the ones column. Now subtract $17 - 8$ in the ones column.

$$\begin{array}{r} 417 \\ \cancel{5}7 \\ - 38 \\ \hline 9 \end{array}$$

2. Regrouping 1 ten from the tens column left 4 tens. Subtract $4 - 3$ and write the result in the tens column of your answer. Check: $19 + 38 = 57$.

$$\begin{array}{r} 417 \\ \cancel{5}7 \\ - 38 \\ \hline 19 \end{array}$$

Sue is **19 years older** than Kathy.

Example 3: Find the difference between 205 and 67.

1. Subtract. Start with the ones column. Since 5 is less than the number being subtracted (7), regroup. Since there are 0 tens in the tens column, regroup 1 hundred from the hundreds column. From 10 tens, regroup 1 ten to the ones column. Now subtract $15 - 7$ in the ones column.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 8 \end{array}$$

2. Regrouping 1 ten from the tens column left 9 tens. Subtract $9 - 6$, and write the result in the tens column of your answer.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 38 \end{array}$$

3. Regrouping 1 hundred from the hundreds column left 1 hundred. Subtract the hundreds column: $1 - 0$. Check: $138 + 67 = 205$.

$$\begin{array}{r} 9 \\ 1 \cancel{10} 15 \\ \cancel{2} \cancel{0} \cancel{5} \\ - 67 \\ \hline 138 \end{array}$$

Practice 2.1

A. Solve.

$$\begin{array}{r} 1. \quad 54 \\ + 23 \\ \hline \end{array}$$

$$\begin{array}{r} 3. \quad 73 \\ - 21 \\ \hline \end{array}$$

$$\begin{array}{r} 5. \quad 105 \\ + 85 \\ \hline \end{array}$$

$$\begin{array}{r} 7. \quad 100 \\ - 57 \\ \hline \end{array}$$

$$\begin{array}{r} 2. \quad 46 \\ + 54 \\ \hline \end{array}$$

$$\begin{array}{r} 4. \quad 55 \\ - 19 \\ \hline \end{array}$$

$$\begin{array}{r} 6. \quad 2386 \\ + 1692 \\ \hline \end{array}$$

$$\begin{array}{r} 8. \quad 2500 \\ - 383 \\ \hline \end{array}$$

B. Rewrite the problems in columns before solving.

$$9. 20 + 12 + 33 =$$

$$13. 144 + 238 + 101 =$$

$$17. 10,326 + 982 =$$

$$10. 245 - 131 =$$

$$14. 545 - 89 =$$

$$18. 15,890 - 705 =$$

$$11. 30 + 75 + 75 =$$

$$15. 2095 + 324 =$$

$$19. 108,755 + 22,442 =$$

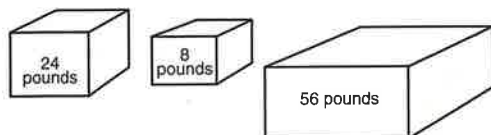
$$12. 378 - 85 =$$

$$16. 1250 - 350 =$$

$$20. 44,789 - 13,890 =$$

C. Choose the one best answer to each question.

21. What is the total weight, in pounds, of the boxes below?



- A. 78
- B. 88
- C. 150
- D. 160

22. Celia's share for lunch is \$7. If she pays with a \$20 bill, how much change should she get?

- A. \$3
- B. \$7
- C. \$13
- D. \$27

Answers and explanations begin on page 661.

Multiplication and Division

You **multiply** to combine the same amount multiple times. For example, instead of adding $24 + 24 + 24$, you could multiply 24 by 3. If a problem asks you to find the **product** of two or more numbers, you should multiply.

Example 4: Find the product of 24 and 63.

1. Align place values as you rewrite the problem in a column.
Multiply the ones place of the top number by the ones place of the bottom number: $4 \times 3 = 12$. Write the 2 in the ones place in the first partial product. Regroup the 1 ten.

$$\begin{array}{r} 1 \\ 24 \\ \times 63 \\ \hline 2 \end{array}$$

2. Multiply the tens place in the top number by 3: $2 \times 3 = 6$.
Then add the regrouped amount: $6 + 1 = 7$. Write the 7 in the tens place in the partial product.

$$\begin{array}{r} 1 \\ 24 \\ \times 63 \\ \hline 72 \end{array}$$

3. Now multiply by the tens place of 63. Write a **placeholder** 0 in the ones place in the second partial product, since you're really multiplying by 60. Then multiply the ones place of the top number by 6: $4 \times 6 = 24$. Write 4 in the partial product and regroup the 2. Multiply $2 \times 6 = 12$. Add the regrouped 2: $12 + 2 = 14$.

$$\begin{array}{r} 2 \\ 24 \\ \times 63 \\ \hline 72 \\ + 1440 \\ \hline 1512 \end{array}$$

4. Add the partial products to find the total product:
 $72 + 1440 = 1512$.

To **divide** means to find how many equal parts an amount can be divided into. The amount being divided is called the **dividend**. The number you are dividing by is the **divisor**, and the answer to a division problem is the **quotient**.

Example 5: At a garage sale, 3 children sold their old toys for a total of \$54. If they share the money equally, how much money should each child receive?

1. Divide the total amount (\$54) by the number of ways the money is to be split (3). Work from left to right. How many times does 3 go into 5? Write the answer 1 directly above the 5 in the dividend. Since $3 \times 1 = 3$, subtract $5 - 3 = 2$.

$$\begin{array}{r} 1 \\ 3 \overline{) \$54} \\ \underline{-3} \\ 2 \end{array}$$

2. Continue dividing. Bring down the 4 from the ones place in the dividend. How many times does 3 go into 24? Write the answer 8 directly above the 4 in the dividend. Since $3 \times 8 = 24$, subtract $24 - 24 = 0$.
Each child should receive \$18.

$$\begin{array}{r} 18 \\ 3 \overline{) \$54} \\ \underline{-3} \\ 24 \\ \underline{-24} \\ 0 \end{array}$$

Example 6: Divide $1006 \div 4$.

1. Divide the total amount (1006) by 4. Work from left to right. Since 4 doesn't divide into 1, use the next place value in the dividend. How many times does 4 go into 10? Write the answer 2 directly above the first 0 in the dividend. Since $4 \times 2 = 8$, subtract $10 - 8 = 2$.

$$\begin{array}{r} 2 \\ 4 \overline{) 1006} \\ \underline{-8} \\ 2 \end{array}$$